

Albert Ankomah Dodoo

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SUMMARY

I am a Machine Learning researcher with expertise in adaptive learning, pattern recognition, swarm intelligence, and computer vision. I am interested in exploring Adaptive Multimodal Learning, Concept Drift Detection, and Trustworthy AI. I have mentored undergraduate students interested in machine learning research and its application in public health and security.

EDUCATION

University of Ghana	Accra, Ghana
MSc. Computer Science	Sep 2024 – Sep 2025
Thesis: Adaptive Memory-Based Cuckoo Search for High-Dimensional Feature Optimization in Non-Stationary Fraud Detection Systems.	
University of Ghana	Accra, Ghana
BSc. Computer Science	Aug 2019 – Sep 2023
Thesis: Resampled Correlation-Based Feature Descriptors: A Novel Approach to Enhancing Malware Detection Capabilities.	

WORK EXPERIENCE

University of Ghana Department of Computer Science	Accra, Ghana
<u>Graduate Teaching Assistant</u>	Oct 2025 – Present
- Support academic writing by converting thesis documents into publishable research articles. - Organise and deliver biweekly LaTeX workshops for over 50 undergraduate and graduate final-year students. - Assist in grading mid-semester assessments and final examinations for 120 graduate students. - Conduct tutorial sessions in Advanced Data Structures and Algorithms, Intelligent Systems, and Natural Language Processing for 120 students.	
Data Intelligence and Swarm Analytics Laboratory (DISAL)	Accra, Ghana
<u>Teaching and Research Assistant</u>	Aug 2022 - Present
- Mentor 11 undergraduate in machine learning research, guiding projects through weekly meetings. - Conduct tutorial session in python programming and best literature review practices for members of the lab. - Integrated EDDM and ADWIN for real-time detection of gradual and abrupt concept drift, enabling model adaptation to changing patterns in fraud data. - Developed an online LSTM framework for continuous adaptation to evolving transaction data streams. - Collaborated with two undergraduate students on conducting research on Vision Transformer-based COVID-19 detection using chest CT scans. - Collaborated with two colleague researchers on a Swin-LSTM IoT intrusion detection pipeline, achieving >90% precision, recall, and F1-score across benchmark datasets. - Conducted a systematic literature review on network security and privacy using the PRISMA framework to explore trends since 2020.	
Knoxsi Ghana Limited	Accra, Ghana
<u>Data Scientist (Contract)</u>	Oct 2025 – Apr 2026
- Designed and optimized complex SQL queries (including multi-table JOINS) to transform raw health data into structured datasets for analysis and reporting. - Analyzed patient vital datasets to generate insights that support clinical and business decision-making. - Developed interactive dashboards and visualizations using Apache Superset and Tableau to communicate trends, anomalies, and key performance indicators to stakeholders.	
Space Copy Inc.	Delaware, United States
<u>STEM Research Intern</u>	Jun 2025 – Sep 2025

- Reviewed 50+ peer-reviewed papers on on-device AI, runtime optimization, and explainability for autonomous edge systems.
- Assessed embedded vision and sensor fusion methods in lunar conditions, focusing on optimization trade-offs between accuracy and efficiency.
- Supported grant writing for the NASA CAN on SLM Additive Manufacturing Dual Technologies (MSFC).

University of Ghana Department of Computer Science

Accra, Ghana

Teaching Assistant

Sep 2023 – Oct 2024

- Led tutorial sessions for 70 students on subjects like Machine Learning, Numerical Methods, and Statistical Models for Data Science to improve their understanding.
- Demonstrated practical applications of statistical concepts such as standardization, z-score, and measures of central tendency through Python code implementations during lectures.
- Designed quizzes for 70 students and provided feedback to enhance students' understanding.

PUBLICATIONS AND PREPRINTS

- Appati, K.J., Ziamah, B., Akrofi, H.A., & **Dodoo, A.A.** SARS Detection in Chest CT-Scan Images Using Bootstrapped ViT-B16 Model. Springer, Iran Journal of Computer Science (2025). doi.org/10.1007/s42044-025-00231-1
- **Dodoo, A.A.**, Koranteng, M., Amissah, D.K., & Appati, K.J. Resampled Correlation-Based Feature Descriptors: A Novel Approach to Enhancing Malware Detection Capabilities. Preprint at *Research Square* (2023). doi.org/10.21203/rs.3.rs-4219089/v1
- Quaicoo, R., Acheampong, R., Gyamenah, P., **Dodoo, A.A.**, Soli, M.A.T., & Appati, K.J. Adapting Triple BigGAN for Image Detection Tasks: Challenges and Opportunities. Preprint at *Research Square* (2023). doi.org/10.21203/rs.3.rs-4262097/v1

ACADEMIC AND RESEARCH ENGAGEMENT

Deep Learning Indaba

Lagos, Nigeria

General Attendee

Aug 2026

- Presented my master's research on adaptive feature optimization for non-stationary fraud detection.
- Participated in a Mila-Quebec AI Institute workshop on articulating research interests and identifying relevant faculty.
- Completed a hands-on workshop on understanding the maths behind neural networks using Google Colab.

African Computer Vision Summer School (ACVSS)

Accra, Ghana

Participant & Local Organizer

July 2026

- Gained practical foundations in 3D computer vision, including camera geometry, calibration, feature correspondences, and Structure-from-Motion, taught by Prof. Ronald Clark of University of Oxford.
- Built a conceptual understanding of modern generative AI, covering VAEs, GANs, autoregressive models, and diffusion-based generation with Prof. Deepti Ghadiyaram of Boston University.
- Collaborated on a medical imaging hackathon developing tumor-aware MRI super-resolution with segmentation and Monte Carlo Dropout-based uncertainty estimation.
- Engaged with Max Welling (Cusp AI), Sekou Remy (Google), and Amr Kahalifa (Google DeepMind) through keynote sessions on emerging research directions and applications of AI.

AWARDS, RECOGNITIONS, AND CERTIFICATIONS

- Best Poster Presentation Award, Deep Learning Indaba, Lagos-Nigeria (August 2026)
- Overall Best Graduating Graduate Student at the Department of Computer Science (December 2025).
- DataCamp Donates X DataGlobal Hub Scholarship (October 2025)
- Overall Best Research Paper Award at the University of Ghana Postgraduate Research Workshop (July 2025)

TECHNICAL SKILLS

Programming Languages: Python | SAS | R | MATLAB | SQL

Visualization Tools: Apache Superset | Power BI

Productivity Tools: Git | Microsoft Office Suites | LaTeX